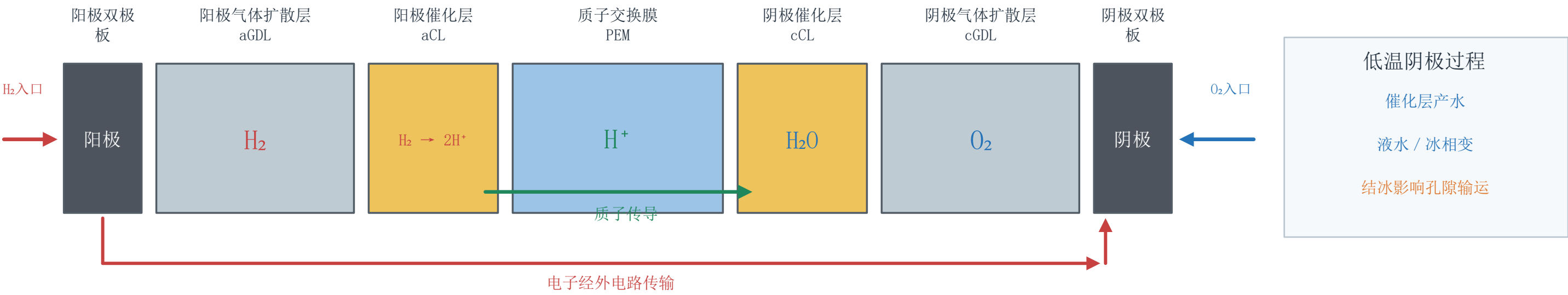
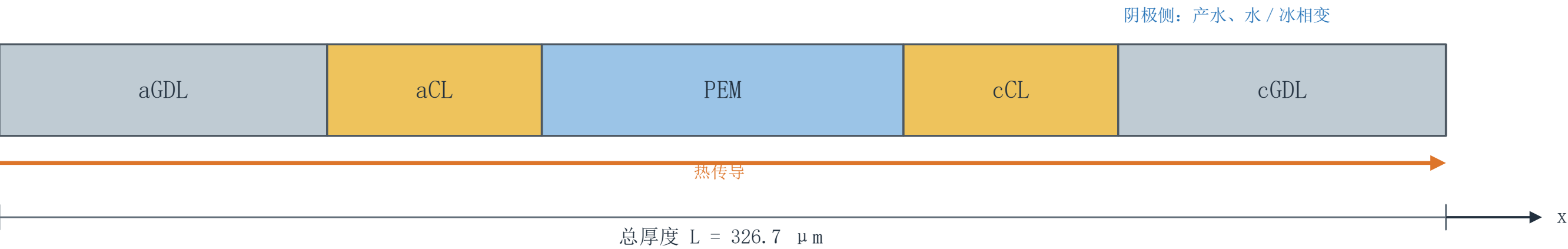


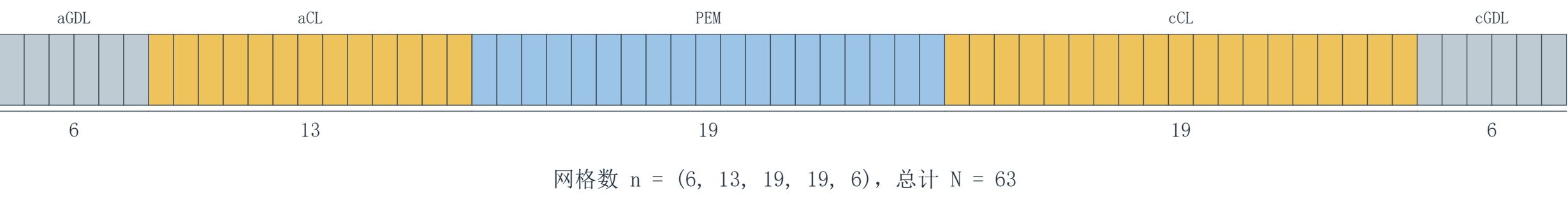
(a) 单电池分层结构与主要运输过程



(b) 一维求解区域（沿厚度 x 方向）



(c) 有限体积离散



状态量：温度、氢氧浓度、液水量、冰量、阴极催化层含水量